GENERAL SIR JOHN KOTELAWALA DEFENSE UNIVERSITY

BACHELOR OF SCIENCE IN APPLIED DATA SCIENCE COMMUNICATION

ASSIGNMENT - 01

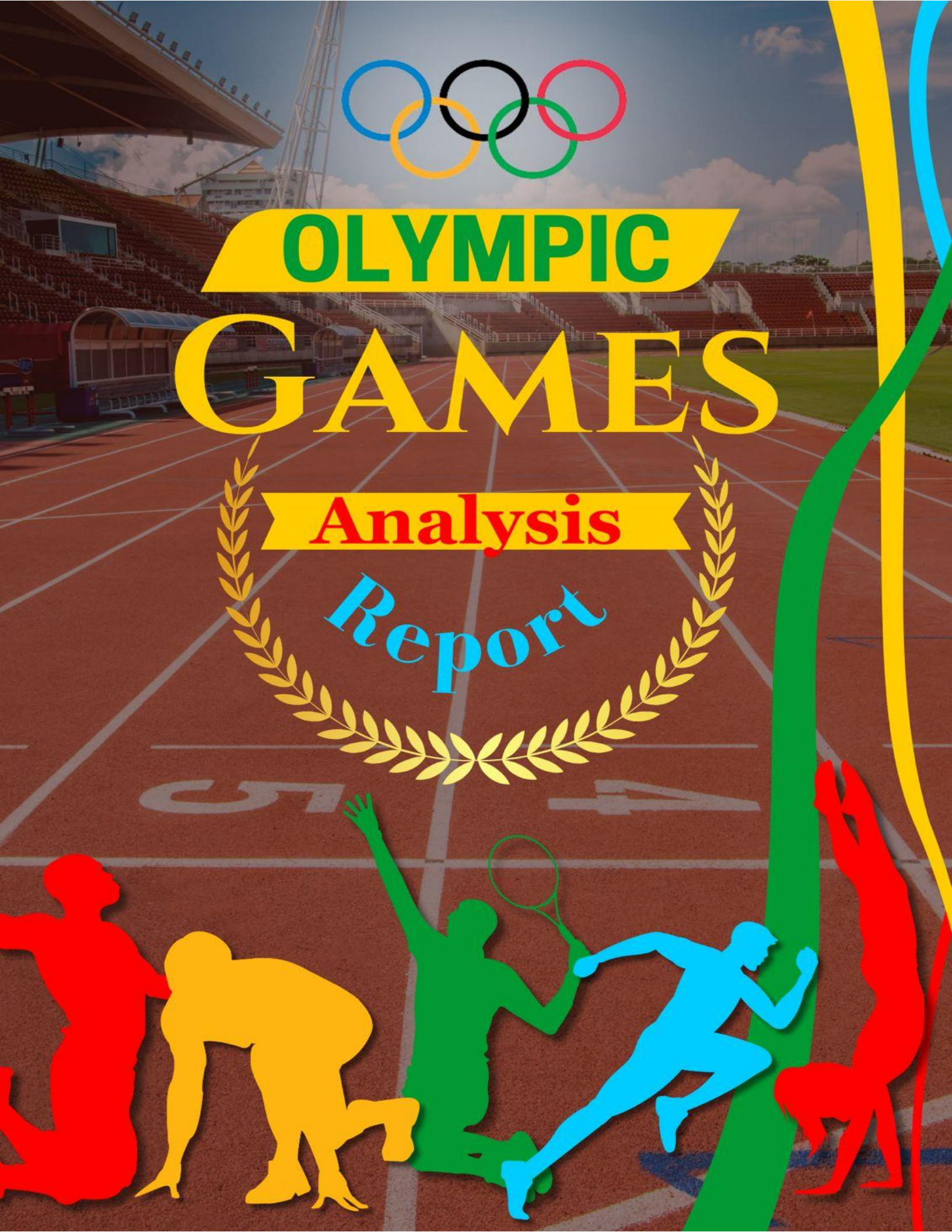
D/ADC/24/0026 – MWRS De Silva

D/ADC/24/0018 – KAUS Perera

D/ADC/24/0019 – DD Weerasuriya

OLYMPIC GAMES

Analysis Report



Olympic Games Analysis Report

The Olympic games, the epitome of international sporting competition, have produced vast data throughout history. To understand the trends and patterns of this dataset, we created an interactive dashboard to highlight the key aspects of Olympic performance and participation. Our dashboard contains data from both the summer and winter Olympics covering decades of various sports. We inspected and visualized several factors like medal distribution, athlete demographics and participation. This report analyzes the data to find important insights, patterns and trends throughout the course of the Olympics and the performance of each nation.

Data Extraction & Cleaning

The datasets that were used to visualize the dashboard in Power BI was obtained from the all-time Olympic Games medal table from Wikipedia - this table contains details regarding the dispersion of summer and winter Olympic medals between the participated nations and Olympic history: longitudinal data table from figshare – this table contains details of sports and athletes over the years.

The dataset from Wikipedia was extracted to excel through the get data function where we linked the table on Wikipedia to excel where we removed missing values and duplicates from excel itself as we were doing the visualizations we ran into a problem where the world map didn't recognize the country names from the team column to solve this, we duplicated the column by example and split it by delimiter.

The dataset from figshare was loaded to R studio and there we cleaned the data and removed null values in the medal list as we only wanted to consider participants who have won medals. Afterwards we removed columns unnecessary for our analysis such as the personal details of the athlete. Then we loaded it to power bi after exporting it from r as an .xlsx file. After which we sorted the athletes by country and then by year. After this when we tried to make a relationship between the NOCs in the 2 groups it didn't happen so to clear the data better again split the NOC column after the 1st 3 characters and deleted the second column from the Wikipedia dataset, so we were able to make a relationship.

User Interface & Functionality

The dashboard has a user-friendly interface for easy navigation between pages. The users can personalize the view of the data through the country and year filters that are accessible through the menu. Each visualization on the dashboard includes a "show details" icon that allows users to click and generate a pop-up window for a closer breakdown of the graph. It is through the combination of navigation options, filtering capabilities, and detailed view functionality that a user will cover and analyze efficiently the data presented on the dashboard.

Trends & Patterns

As per our visualizations, we can initially observe that there is a combined total of 20281 total medals across all Olympic games, with 6631 gold, 6598 silver and 7052 bronze medals. combined total of gold medals, 6598 silver medals and 7052 bronze medals. The total number of summer medals account for 16781 and the total of winter Olympic medals are 3500.

Looking at our 100% stacked bar chart representing the total medals won by each country in summer vs winter Olympics, we can see that majority of the countries have a higher medal count for summer Olympics. However, it is very apparent that countries experiencing extreme winter conditions such as Norway, Liechtenstein, Austria have a higher proportion of winter Olympic medals.

The 100% stacked bar chart representing the distribution of gold, silver and bronze medals clearly show how much medals each country has obtained and how much of it are gold, silver and bronze respectively.

The line chart illustrates how the medal count increase throughout the years and the clear difference between the number of medals in summer and winter Olympics. This trend shows the expansion of new games over time.

The 100% stacked column chart representing count of medal by sport and sex reveals an interesting pattern. It shows that medals for more sports are held by men. However, in sports like swimming, handball, volleyball, cross country skiing, badminton, table tennis etc. women have earned equal or more medals. The chart also presents which sports are male dominated and female dominated.

According to the line chart representing count of sport by year and sex provides insights like even though initially there is a major gap between the number of male and female participants starting from 1908, we see a slow increment of female participants and a major leap from 1970 to 1976 where the number of female participants almost doubles. After which there is a steady increase in the number of female participants and in 2016 the difference between male and female participants is very narrow.

The top three sports with the highest number of medals are athletics, swimming and rowing respectively.

As per the scatter plots, USA has the highest number of summer Olympic medals with only a few number of countries that having participated for more than 25 summer Olympics. There is a big difference between the total number of medals won by the USA compared to other countries. For winter Olympics Norway has the highest amount of winter Olympic medals and there is a lower disparity distribution between the number of medals won by other countries compared to Norway.

Finally, the world map visualizes all the countries that have won medals in the Olympics while revealing the regional strengths and weaknesses in performance.

Key overall Trends

- Medal distribution generally shows a pyramid structure with more bronze than silver and more silver than gold.
- The number of medals has increased in growth from about 1980.
- Summer Olympics dominate in terms of participation and medals
- Some countries specialize in either summer or winter games.

Conclusion

The analysis of Olympic games has given valuable insights in showing medal distributions, trends in participation and gender dynamics. The analysis reveals interesting patterns like how different countries dominate summer and winter Olympics and the connection between geographical location of these countries, how the number of events increase and with that the gender disparity decreases over time. The overall report still captures the evolution of the Olympics and what brings success to different nations.

References

All-time Olympic Games medal table n.d - https://en.wikipedia.org/wiki/All-time_Olympic_Games_medal_table

Olympic history: longitudinal data (2018) - Olympic history: longitudinal data scraped from www.sports-reference.com (figshare.com)

Jason Davidson (2022) Creating a Pop-Out Navigation and Filter Bar Using Power BI Bookmarks – Available at: https://www.youtube.com/watch?v=SCI9iKl4vjU

300 Second Tutorials (2022) How to create a popup in Power BI in 300 seconds – Available at: https://www.youtube.com/watch?v=GBu-W_kCHRA